Training Register Demo - High-Level Design

SYN-TRN-HLD-001 | 1.0.0 | SYNTHETIC TEST FIXTURE

Repository: https://github.com/rdhepe/testrepo

Purpose and ownership

Synthetic input for UC3 assessment scenarios, including UC3-WF-01 and TC-DS-01. This is not a PETRONAS production application, approved enterprise design, or compliance certification.

Training Register Demo (SYN-TRN-001), version 1.0.0, registers employees for training and tracks course completion. Business owner: Synthetic Learning Operations. Technical owner: Synthetic Demo Engineering. Contact: training.owner@example.test.

Components, deployment and data flow

Component COMP-HTTP-01: A local browser or HTTP client calls one Node.js HTTP service implemented in src/server.js. GET / serves public/index.html and GET /app.js serves public/app.js. The registration frontend lists and submits enrolments through /api/enrolments. The service binds to 127.0.0.1, port 3100 by default; PORT can override the port. No cloud resource, identity provider, or external integration is specified.

Component COMP-DB-01: The application persists enrolment records in a PostgreSQL database. POST /api/enrolments writes each accepted enrolment to PostgreSQL, and GET /api/enrolments reads the stored records. Accepted enrolments remain available after the Node.js service restarts.

Data flow: local HTTP client -> Node.js HTTP service -> PostgreSQL enrolment database -> JSON response to client. fixtures/application.json supplies application metadata and two synthetic seed records. No infrastructure-as-code deployment or deployed resource inventory is supplied; this document alone does not verify deployed state.

API and data contract

GET /health returns health and application ID. GET /api/application returns metadata and declared data fields. GET /api/enrolments lists records. POST /api/enrolments validates and adds a record, returning HTTP 201 and a generated id UUID. Seed records have no generated id.

POST accepts only employeeId, employeeName, workEmail, courseCode and completionStatus. All are required non-empty strings with a maximum length of 160 characters. workEmail must have email syntax. completionStatus is enrolled or completed. Unknown fields are rejected. The request size limit is 16 KiB.

Invalid input returns HTTP 400, unsupported content type 415, oversized input 413, and unknown routes 404. No passwords, national IDs, IP addresses, or authentication tokens are stored in enrolment records.

Trust boundary and limitations

The local machine is the deployment boundary. Loopback binding limits network exposure but does not authenticate local clients. HTTP is not TLS protected. There is no authentication, authorization, MFA, audit trail, encryption-at-rest mechanism, automated retention, backup or recovery.

Use synthetic records only. Do not expose the service publicly or store real personal data. These limitations are intentional and are not claims of passing other UC3 privacy or security controls.